# EDGE Computing

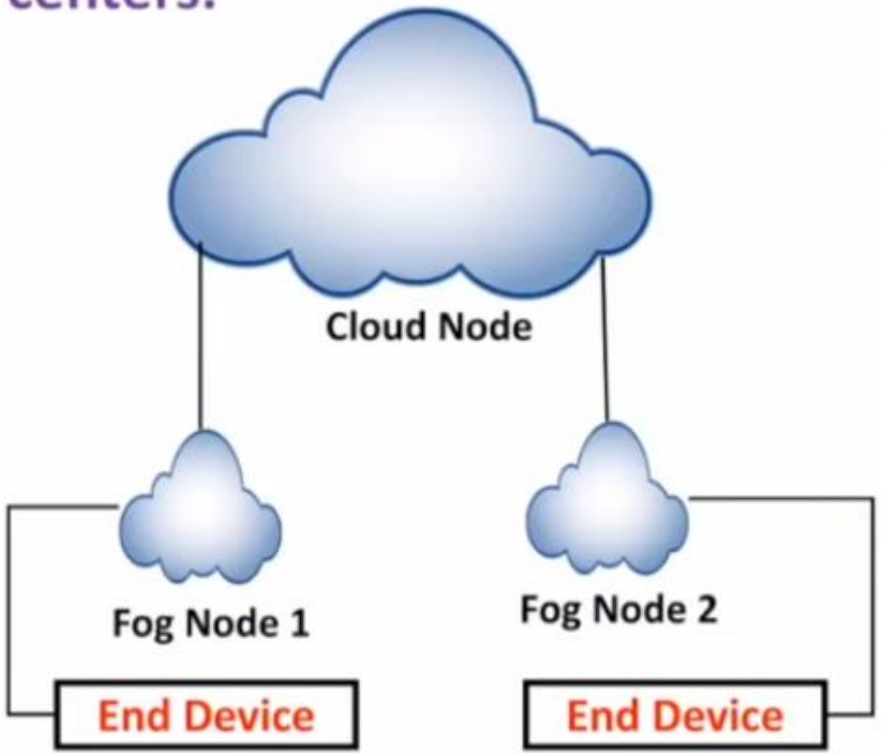
- Edge computing is a computing architecture that aims to bring computing closer to the source of data.
- It is based on the idea of processing data at the edge of the network, as opposed to in the cloud or in a centralized data center. The idea behind edge computing is to reduce the amount of data that needs to be sent to the cloud or a central server for processing, thereby reducing network latency and improving overall system performance.

# FOG Computing

- Fog computing is a distributed computing model that is designed to complement edge computing. It extends the capabilities of edge computing by providing a layer of computing infrastructure between the edge devices and the cloud. This infrastructure is called the fog layer, and it provides additional computing resources and services to edge devices.

# Fog Computing

- Fog Computing is the term coined by Cisco that refers to extending cloud computing to an edge of the enterprise's network.
- Thus, it is also known as Edge Computing or Fogging.
- It facilitates the operation of computing, storage, and networking services between end devices and computing data centers.




Fog Computing Architecture

# Basics of Fog Computing

- The devices comprising the fog infrastructure are known as fog nodes.
- In fog computing, all the storage capabilities, computation capabilities, and data along with the applications are placed between the cloud and the physical host.
- All these functionalities are placed more towards the host. This makes processing faster as it is done almost at the place where data is created.
- It improves the efficiency of the system and is also used to ensure increased security.

# When to use Fog Computing

- **Fog Computing can be used in the following scenarios:**

- It is used when only selected data is required to send to the cloud. This selected data is chosen for long-term storage and is less frequently accessed by the host.
- It is used when the data should be analyzed within a fraction of seconds i.e Latency should be low.
- It is used whenever a large number of services need to be provided over a large area at different geographical locations.
- Devices that are subjected to rigorous computations and processing must use fog computing.
- Real-world examples where fog computing is used are in IoT devices (eg. Car-to-Car Consortium, Europe), Devices with Sensors, Cameras (IIoT-Industrial Internet of Things), etc.

# Benefits of fog computing

- Reduced usage of cloud resources.
- Faster and real time response.
- No communication gap.
- It provides better privacy as industries can perform analysis on their data locally.

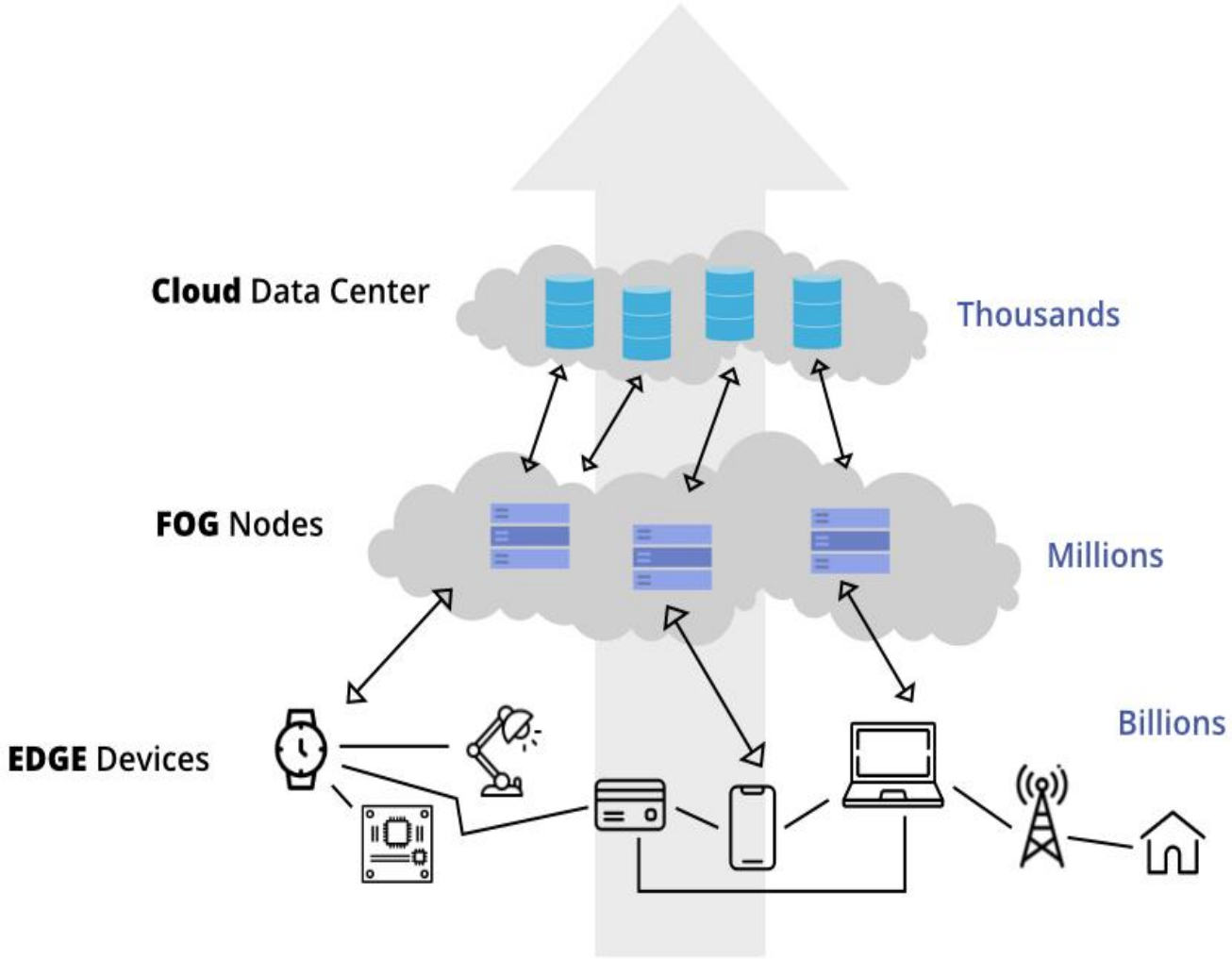
**Disadvantages:**

- Additional resources are required.

# Cloud Computing

- Cloud computing is a centralized model where data is stored, processed, and accessed from a remote data center.
- Cloud computing suffers from higher latency than edge computing and cloud computing.
- Cloud computing is a highly scalable model that can handle a vast amount of data processing and storage requirements.
- Cloud computing has advanced security measures in place to secure data in the cloud

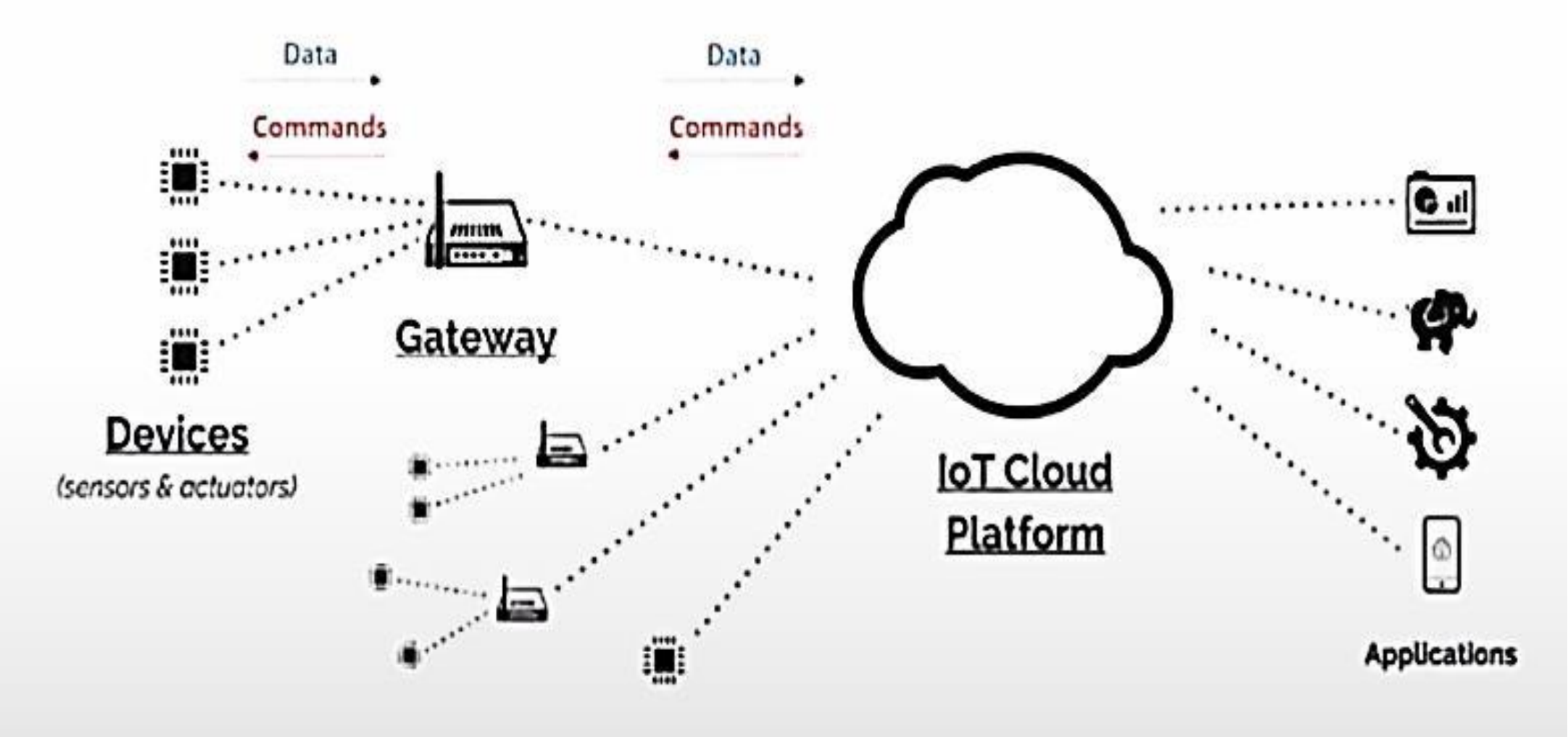
# FOG vs CLOUD


**Cloud** Data Center
Thousands
**FOG** Nodes
Millions
**EDGE** Devices
Billions

# Sensor Cloud

- We can store huge amount of data.
- We can monitor and control sensor data.
- These type of structure is used when data is required for long term.
- The data can be monitor through API(application peripheral interface) it could be any app or website.
- The sensor cloud collects information from multiple sensor network.
- So, It can gather, access, process, visualize, analyse, store as well as they can share the data.
- This can also be helpful in patient monitoring system where doctor can fetch multiple patients data from remote location.

# Sensor Cloud Architecture



# Advantages of Sensor Cloud

Cloud computing is very encouraging solution for Sensor-Cloud infrastructure due to several reasons like the agility, reliability, portability, real-time, flexibility, and so forth.

(1)**Analysis.** The integration of huge accumulated sensor data from several sensor networks and the cloud computing model make it attractive for various kinds of analyses required by users through provisioning of the scalable processing power.

(2) **Scalability.** Sensor-Cloud enables the earlier sensor networks to scale on very large size because of the large routing architecture of cloud.
It means that as the need for resources increases, organizations can scale or add the extra services from cloud computing vendors without having to invest heavily for these additional hardware resources.

(3) **Collaboration**. Sensor-Cloud enables the huge sensor data to be shared by different groups of consumers through collaboration of various physical sensor networks. It eases the collaboration among several users and applications for huge data sharing on the cloud.

(4) **Visualization.** Sensor-Cloud platform provide a visualizationAPI to be used for representing the diagrams with the stored and retrieved sensor data from several device assets. Through the visualization tools, users can predict the possible future trends that have to be incurred.

(5) **Free Provisioning of Increased Data storage and Processing Power.** It provides free data storage and organizations may put their data rather than putting onto private computer systems without hassle. It provides enormous processing facility and storage resources to handle data of large-scale applications.

(6) **Dynamic Provisioning of Services.** Users of Sensor-Cloud can access their relevant information from wherever they want and whenever they need rather than being stick to their desks.

(7)**Multitenancy**. The number of services from several service providers can be integrated easily through cloud and Internet for numerous service innovations to meet user's demand.Sensor-Cloud allows the accessibility to several numbers of data centers placed anywhere on the network world.

(8) **Automation.** Automation played a vital role in provisioning of Sensor-Cloud computing services.Automation of services improved the delivery time to a great extent.

(9) **Flexibility.** Sensor-Cloud provides more flexibility to its users than the past computingmethods. It provides flexibility to use random applications in any number of times and allows sharing of sensor resources under flexible usage environment.

(10) **Agility of Services.** Sensor-Cloud provides agile services and the users can provision the expensive technological infrastructure resources with less cost .The integration of wireless sensor networks with cloud allows the high-speed processing of data using immense processing capability of cloud.

# Examples of sensor cloud:

1. **Smart Cities (e.g., Singapore's Smart Nation Initiative)**

- **Sensors:** Deployed across the city to monitor traffic flow, air quality, noise levels, and other urban data.
- **Data Processing:** Edge devices handle immediate data, while cloud platforms process historical data to identify trends and optimize urban services.
- **Applications:** Improved transportation systems, better waste management, optimized energy usage, and enhanced public safety initiatives.

2. **Industrial Automation (e.g., Siemens MindSphere)**

- **Sensors:** Installed on manufacturing equipment to collect real-time data on performance, temperature, vibrations, and other operational parameters.
- **Data Processing:** Data is first processed locally by edge devices for immediate safety and control. Then, it's sent to the cloud for long-term storage and analysis.
- **Applications:** Enabling predictive maintenance (e.g., fixing a part before it breaks), improving production efficiency, and optimizing overall industrial processes.